ALEXANDER DSOUZA

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PROFESSIONAL SUMMARY

Principal Engineer with 15+ years building and scaling software systems, specializing in Python and DevOps. Proven track record architecting distributed systems and leading cross-functional teams. AI-first engineer skilled in prompt engineering and LLM integration. Recognized Top External Contributor to Mozilla CI.

TECHNICAL SKILLS

- **Backend:** Python, Django, FastAPI, Django REST Framework (DRF), Pydantic, Golang, Rust, Node.js, SQL
- **Frontend:** React, Redux, React Router, Vite, Material UI (MUI), TypeScript, Streamlit, PyDeck
- **Data Engineering & Cloud:** Apache Beam, Cloud Dataflow, GCP Pub/Sub, BigQuery GIS, Cloud Run, Cloud Storage, PostgreSQL, AlloyDB
- **Infrastructure & DevOps:** Kubernetes (k3s/GKE), Docker, AWS, Ansible, Helm, CI/CD (GitHub Actions, GitLab)
- **Observability:** Prometheus, Grafana, Jaeger, Loki, OpenTelemetry, Google Cloud Monitoring, New Relic, Snyk
- **Core Expertise:** Microservices, SDN, Distributed Systems, API Design (Protobuf/JSON), Message Queues (RabbitMQ), SOC 2 Compliance
- **AI/ML:** Prompt Engineering, LLM Integration, Daily GenAI-Assisted Development (Cursor, Aider, Jules, Copilot), Machine Learning Fundamentals

PROFESSIONAL EXPERIENCE

Independent Contributor — Open Source & Backend Engineering Practice
Oct 2025 – Present

Self-Directed | Pune, India

- Completed the AWS AI/ML Scholar Challenge 2026 (Udacity); recognized on planet.mozilla.org throughout 2026 for Treeherder/PerfCompare contributions.
- Deepened expertise in FastAPI and Django, and built hands-on proficiency in React, TypeScript, and Redux via component and state-management work on PerfCompare.
- Advanced PostgreSQL/Redis via query optimization, indexing, and scaling patterns.

Principal Software Engineer
Apr 2018 – Oct 2025

Jeavio India Pvt Ltd | Pune, India

Promoted twice internally over 7.5 years: Senior QA Engineer → Senior Software Engineer → Principal Engineer.

- Architected and led cross-functional teams delivering SDN and WiFi collectors (async, multithreaded Python, high-throughput JSON processing) for IBM SevOne NMS across multiple releases, for enterprise customers including BT; owned sprint planning and daily scrums.
- Introduced structured on-call rotation and documented RCA practices to prevent burnout and speed up fixes; drove a Shift-Left QA approach that increased unit test coverage.
- Singlehandedly architected and maintained CI/CD pipelines and installers across all collector teams: designed multi-repo GitLab pipelines (migrated to GitHub Actions in the 2023 IBM transition) integrated with vSphere, JFrog, and Slack, architected Docker containerization for the collectors, and authored the Ansible playbooks for end-to-end installation.
- Delivered the Versa SD-WAN collector as Senior Software Engineer, then oversaw the migration/rewrite of the SDN and WiFi collectors from Python to Golang and led the IBM transition; later led full-stack design (DRF API layer, frontend architecture) for a global dairy enterprise and its SOC 2 AWS upgrade.

Quality & Test Engineering — Amazon, Honeywell, Avaya, Wipro
Feb 2008 – Feb 2018

- **Amazon (QA Engineer II, Global Address Management):** built the Global Address Model, led Source Data Quality validation for Gold Address Data providers (GB/US), and developed the Protobuf-to-JSON SerDe backbone for test-system data storage; published a tutorial on it (medium.com/@moijes12).
- **Honeywell (Senior Test Engineer):** architected a custom performance measurement test framework in Python; led mobile automation feasibility, selecting Appium.
- **Avaya (Software Test Engineer):** Test Lead for the One Touch Video and Social Media Manager platforms; administered VMware ESX test environments and automated database tasks.
- **Wipro (Project Engineer):** handled end-to-end testing and automation for Nokia Base Station Controller GPRS systems.

OPEN SOURCE & KEY ACHIEVEMENTS

- Mozilla CI Top Contributor: merged 11+ PRs to Treeherder and PerfCompare, working with RabbitMQ-based Pulse task/job queueing, React, React Router, and TypeScript.
- Legacy Resolution: resolved 21+ legacy Mozilla issues in Treeherder. Hackathon Winner: designed an on-demand environment pipeline reducing issue-reproduction setup time. Mentored interns into full-time roles.

EDUCATION

Bachelor of Engineering, Information Technology
Mumbai University (FCRIT)